Hull White Simulation 로직

1. 이론적 기초

HW 1Factor Process를 다음과 같이 가정한다.

$$\mathrm{r_t} = x_t + \phi_t$$

$$\mathrm{dx_t} = -\kappa x_t dt + \sigma_t dW$$

2. 확률과정 $\mathbf{e^{\kappa t} \cdot x_t}$에 대하여

$$\mathrm{d}(\mathrm{e}^{\kappa t} \cdot x_t) = \kappa e^{\kappa t} x dt + e^{\kappa t} dx_t = \kappa e^{\kappa t} x_t dt + e^{\kappa t}[-\kappa x_t dt + \sigma_t dW]$$

$$= \kappa e^{\kappa t} x_t dt + e^{\kappa t}(-\kappa x_t) dt + \sigma_t e^{\kappa t} dW = \sigma_t e^{\kappa t} dW$$

양 변을 T1에서 T2까지 적분하면

$$\mathrm{e}^{\kappa \mathrm{T}_2} \times x_{T_2} - \mathrm{e}^{\kappa \mathrm{T}_1} \times x_{T_1} = \int_{T1}^{T2} \sigma_t e^{\kappa t} dW$$

$$\mathrm{x_{T2}} = \frac{e^{\kappa T_1}}{e^{\kappa T_2}} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW = e^{-\kappa(T_2-T_1)} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW$$

$$r_{\mathrm{T}_2} = e^{-\kappa(T_2-T_1)} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW + \phi_{T_2}$$

$$\mathrm{E}\left(\mathrm{x}_{\mathrm{T}_2} | T_1\right) = e^{-\kappa(T_2-T_1)} x_{T_1}$$

$$\mathrm{V}\left(\mathrm{x}_{\mathrm{T}_2} | T_1\right) = V\left(\int_{T_1}^{T_2} \sigma_t e^{\kappa(t-T_2)}\, dW | T_1\right) = \mathrm{E}\left(\left(\int_{T_1}^{T_2} \sigma_t e^{\kappa(t-T_2)}\, dW\right)^2 \middle| T_1\right)$$

$$= \mathrm{E}\left(\int_{T_1}^{T_2} \sigma_t^2 e^{2\kappa(t-T_2)}\, dt\right) = \int_{T_1}^{T_2} \sigma_t^2 e^{-2\kappa(T_2-t)}\, dt$$

3. 따라서, 차기 x(t)은 다음과 같이 계산

$$\mathrm{x(t)|x(s)} \sim N\left(e^{\kappa(t-s)} x(s), \int_s^t \sigma_u^2 e^{-2\kappa(t-u)} du\right)$$

$\mathrm{XA}(\mathrm{t_i}) = e^{-\kappa(t_{i+1}-t_i)}$ 이고, $\mathrm{XV}(\mathrm{t_i}) = \left(\int_{t_i}^{t_{i+1}} e^{-2\kappa(t_{i+1}-\tau)}\sigma^2(\tau)d\tau\right)^{0.5}$

만약 Vol이 Simple Vol이라면,

$$\mathrm{XV}(\mathrm{t_i}) = \left(\int_{t_i}^{t_{i+1}} e^{-2\kappa(t_{i+1}-\tau)}\sigma^2 d\tau\right)^{0.5} = \left(\frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa(t_{i+1}-t_i)}\right)\right)^{0.5}$$

따라서, $\mathbf{x}(\mathbf{t_{i+1}}) = \mathbf{XA}(\mathbf{t_i}) \cdot \boldsymbol{x}(\boldsymbol{t_i}) + \boldsymbol{XV}(\boldsymbol{t_i}) \cdot \boldsymbol{\epsilon_i}$이며 다음과 같이 구현한다.

```
//A_k = exp(-kappa * (t_(k+1) - t_(k))
double XA(
    double kappa,
    double t0,
    double t1
)
{
    double A_k = exp(-kappa * (t1 - t0));
    return A_k;
}

double XV(
    double kappa,
    long NHWVol,
    double* HWVolTerm,
    double* HWVol,
    double t0,
    double t1
)
{
    long i;
    double vol = 0.0;
    double B_k, B_k_Square;
    kappa = max(0.00001, kappa);

    if (NHWVol == 1 || t1 - t0 < 0.25)
    {
        vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, t0);
        B_k_Square = vol * vol * 0.5 / kappa * (1.0 - exp(-2.0 * kappa * (t1 - t0)));
    }
    else
    {
        long NInteg = 10.0;
        double u = t0;
        double du = (t1 - t0) / ((double)NInteg);
        B_k_Square = 0.0;
        for (i = 0; i < NInteg; i++)
        {
            vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, u);
            B_k_Square += vol * vol * exp(-2.0 * kappa * (t1 - u)) * du;
            u = u + du;
        }
    }

    B_k = sqrt(B_k_Square);
    return B_k;
}
```

4. $\boldsymbol{P_{HW}(t,T) = E\left(\mathrm{e}^{-\int_t^T r_u du}\middle| F_t\right) = \mathrm{E}\left(\mathrm{e}^{-\int_t^T (x_u+\phi_u)du}|F_t\right)}$**이다.**

x_t는 정규분포를 따르고, 위 수식은 정규분포 적률생성함수(MGF) 형태이다.

$$\mathrm{E}(\mathrm{e}^{\mathrm{tx}}) = e^{\mu t+\frac{1}{2}(\sigma t)^2}$$

$$\boldsymbol{B(s,t,\kappa)} = \int_{\mathrm{s}}^{\mathrm{t}} XA(u,t,\kappa)du = \int_{\mathrm{s}}^{\mathrm{t}} e^{-\kappa(t-u)}du = \boldsymbol{\frac{1-\mathrm{e}^{-\kappa(\mathrm{t}-\mathrm{s})}}{\kappa}}$$

$$\mathrm{VT}(\mathrm{T}_1,\mathrm{T}_2,\kappa,\{\sigma\}) = \mathrm{V}\left(\int_{T_1}^{T_2} X_t dt\right) = V\left(\int_{T_1}^{T_2}\left(\int_t^{t+dt} \sigma_t e^{-\kappa(dt)}\, dW\right)dt\right)$$

$$\approx V\left(\int_{T_1}^{T_2} \sigma_t \cdot \frac{1-e^{-\kappa(T_2-t)}}{\kappa} dW\right)$$

$$\mathrm{V}\left(\int_{T_1}^{T_2} \sigma_t \cdot \frac{1-e^{-\kappa(T_2-t)}}{\kappa} dW\right) = \int_{T_1}^{T_2} \sigma_t^2 \cdot \left(\frac{1-e^{-\kappa(T_2-t)}}{\kappa}\right)^2 dt$$

$$\boldsymbol{\mathrm{VT}(\mathrm{T}_1,\mathrm{T}_2,\kappa,\{\sigma\}) = \int_{T_1}^{T_2} \sigma_t^2 \cdot \left(\frac{1-e^{-\kappa(T_2-t)}}{\kappa}\right)^2 dt}$$

만약 Vol Term이 1개(Simple Vol일 때) 다음과 같이 정리 가능

$$\int_{T_1}^{T_2} \sigma^2 \cdot \left(\frac{1-e^{-\kappa(T_2-t)}}{\kappa}\right)^2 dt = \int_{T_1}^{T_2} \sigma^2 \cdot \left(\frac{1-2e^{-\kappa(T_2-t)}+e^{-2\kappa(T_2-t)}}{\kappa^2}\right) dt$$

$$= \frac{\sigma^2}{\kappa^2}\int_{T_1}^{T_2}\left(1-2e^{-\kappa(T_2-t)}+e^{-2\kappa(T_2-t)}\right) dt$$

$$\boldsymbol{\mathrm{VT}(\mathrm{T}_1,\mathrm{T}_2,\kappa,\sigma) = \frac{\sigma^2}{\kappa^2}\left(T_2-T_1-\frac{2}{\kappa}\left(1-e^{-\kappa(T_2-T_1)}\right)+\frac{1}{2\kappa}\left(1-e^{-2\kappa(T_2-T_1)}\right)\right)}$$

$$\boldsymbol{P_{HW}(t,T)} = \mathrm{E}\left(\mathrm{e}^{-\int_t^T (x_u+\phi_u)du}|F_t\right) = \boldsymbol{\mathrm{e}^{-x_t\cdot B(t,T,\kappa)+\frac{1}{2}VT(t,T,\kappa)-\phi(T)}}$$

· $\boldsymbol{B(t,T,\kappa)}$**와** $\boldsymbol{\frac{1}{2}VT(t,T,\kappa)}$**은 다음과 같이 구현한다.**

```
double B_s_to_t(
    double kappa,
    double s,
    double t
)
{
    return (1.0 - exp(-kappa * (t - s))) / kappa;
}
```

```
double HullWhiteQVTerm(
    double t,
    double T,
    double kappa,
    long NHWVol,
    double* HWVolTerm,
    double* HWVol
)
{
    long i;
    double vol;
    double RHS = 0.0;

    if (NHWVol == 1 || kappa > 0.1)
    {
        vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, t);
        // ...
        RHS = 0.5 * vol * vol / (kappa * kappa) * (T - t - 2.0 / kappa * (1.0 - exp(-kappa * (T - t))) + 0.5 / kappa * (1.0 - exp(-2.0 * kappa * (T - t))));
    }
    else
    {
        RHS = 0.0;
        double s, ds;
        long NInteg = 10.0;
        double u = t;
        double du = (T - t) / ((double)NInteg);
        for (i = 0; i < NInteg; i++)
        {
            vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, u);
            RHS += 0.5 * vol * vol * B_s_to_t(kappa, u, t) * B_s_to_t(kappa, u, t) * du;
            u = u + du;
        }
        // ...
    }
    return RHS;
}
```

5. **반복되는 연산을 피하기 위해 XA, XV, $B(t,T,\kappa)$, $VT(t,T,\kappa,\{\sigma\})$는 시뮬레이션 전에 미리 Generate 해놓고 epsilon을 시뮬레이션을 통해 산출하여 Short-Rate path 시뮬레이션한다.**

$$\mathbf{x}(\mathbf{t_{i+1}}) = \mathbf{XA}(\boldsymbol{t_i}, \boldsymbol{t_{i+1}}, \boldsymbol{\kappa}) \cdot \boldsymbol{x}(\boldsymbol{t_i}) + \boldsymbol{XV}(\boldsymbol{t_i}, \boldsymbol{t_{i+1}}, \boldsymbol{\kappa}, \{\boldsymbol{\sigma}\}) \cdot \boldsymbol{\epsilon_i}$$

$$\boldsymbol{P_{HW}}(\boldsymbol{t},\boldsymbol{T}) = \mathrm{E}\left(\mathrm{e}^{-\int_t^T (x_u+\phi_u)du} | F_t\right) = \mathbf{e}^{-\boldsymbol{x_t}\cdot\boldsymbol{B}(\boldsymbol{t},\boldsymbol{T},\boldsymbol{\kappa})+\frac{1}{2}\boldsymbol{VT}(\boldsymbol{t},\boldsymbol{T},\boldsymbol{\kappa})}$$

여기서 $\mathbf{XA}, \mathbf{XV}$는 Short Rate의 시뮬레이션 t마다 dt 간격으로 생성

2D Matrix Size = (ShortRate시뮬레이션 커브개수, 시뮬레이션의 날짜개수)

또한 $\boldsymbol{B}(\boldsymbol{t},\boldsymbol{T},\boldsymbol{\kappa})$**와** $\frac{1}{2}\boldsymbol{VT}(\boldsymbol{t},\boldsymbol{T},\boldsymbol{\kappa}\{\boldsymbol{\sigma}\})$**는** 각 Rcv, Pay Leg의 기초금리의 만기까지 사전에 Generate해야함.

$\boldsymbol{B_{Rcv}}(\boldsymbol{t_{Simul}}, \boldsymbol{T_{Rcv}}, \boldsymbol{\kappa})$, $\boldsymbol{B_{Pay}}(\boldsymbol{t_{Simul}}, \boldsymbol{T_{Pay}}, \boldsymbol{\kappa})$ **사이즈와**

$\boldsymbol{VT_{Rcv}(t_{Simul}, T_{Rcv}, \kappa, \{\sigma\}), VT_{Pay}(t_{Simul}, T_{Pay}, \kappa, \{\sigma\})}$ **사이즈는**

3D Matrix Size =

(ShortRate시뮬레이션 커브개수, 시뮬레이션의 날짜개수, Leg별 쿠폰개수)

6. 이후 기초금리를 계산한다.

$$\mathrm{R_{HW}(t,T)} = \frac{1 - P_{HW}(t, T_N)}{\sum_1^N [\Delta T_i \times P_{HW}(t, T_i)]}$$

예를 들어 2년 만기 스왑금리를 기초금리라고 가정한다면,

1년 뒤의 시뮬레이션된 금리 산출과정은 다음과 같다.

$$\mathrm{P_{HW}}(1,1.25) = \frac{P(0,1.25)}{P(0,1)} \exp\left(-r_1 \times B(1,1.25) + \frac{1}{2} VT(1,1.25)\right)$$

$$\mathrm{P_{HW}}(1,5) = \frac{P(0,1.5)}{P(0,1)} \exp\left(-r_1 \times B(1,5) + \frac{1}{2} VT(1,5)\right)$$

$$\mathrm{P_{HW}}(1,1.75) = \frac{P(0,1.75)}{P(0,1)} \exp\left(-r_1 \times B(1,1.75) + \frac{1}{2} VT(1,1.75)\right)$$

$$\mathrm{P_{HW}}(1,2) = \frac{P(0,2)}{P(0,1)} \exp\left(-r_1 \times B(1,2) + \frac{1}{2} VT(1,2)\right)$$

$$\mathrm{P_{HW}}(1,2.25) = \frac{P(0,2.25)}{P(0,1)} \exp\left(-r_1 \times B(1,2.25) + \frac{1}{2} VT(1,2.25)\right)$$

$$\mathrm{P_{HW}}(1,2.5) = \frac{P(0,2.5)}{P(0,1)} \exp\left(-r_1 \times B(1,2.5) + \frac{1}{2} VT(1,2.5)\right)$$

$$\mathrm{P_{HW}}(1,2.75) = \frac{P(0,2.75)}{P(0,1)} \exp\left(-r_1 \times B(1,2.75) + \frac{1}{2} VT(1,2.75)\right)$$

$$\mathrm{P_{HW}}(1,3) = \frac{P(0,3)}{P(0,1)} \exp\left(-r_1 \times B(1,3) + \frac{1}{2} VT(1,3)\right)$$

$$\mathrm{R_{HW}^{Swap}}(1,3) = \frac{1 - P_{HW}(1,3)}{\sum_1^8 [\Delta T_i \times P_{HW}(1, T_i)]}$$

끝.